

Ex. No:6 Use Sleuth Kit to analyze digital evidence

Description

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyze digital evidence:

Step 1: Install Sleuth Kit

1. Download Sleuth Kit:

- Visit the official [Sleuth Kit website](https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v) or https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- and download the latest version for Windows.

2. Install Sleuth Kit:

- Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.

Step 2: Acquire the Disk Image

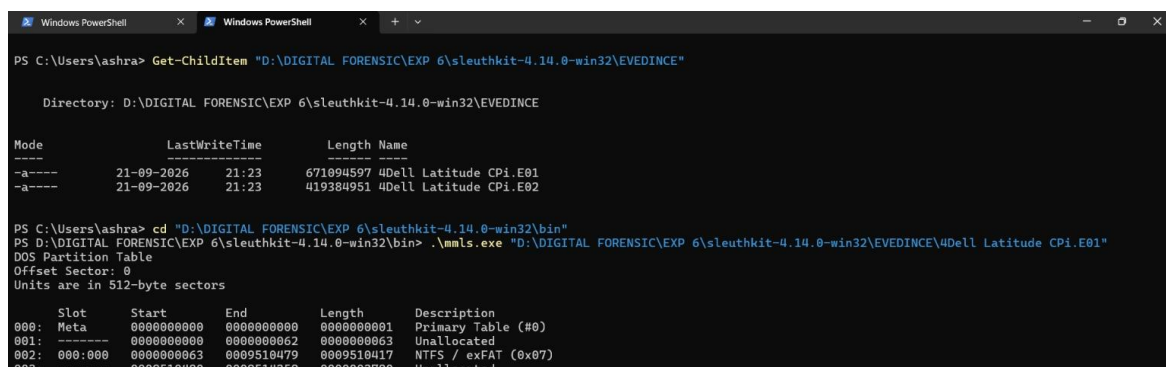
Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

1. Create Disk Image:

- Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device. Ensure the image is in a format supported by Sleuth Kit, such as .dd, .raw, .img, or .E01.
- Download the below files from the google drive

1. 4Dell Latitude CPi.E01

2. 4Dell Latitude CPi.E02



```
PS C:\Users\ashra> Get-ChildItem "D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\EVEDINCE"

Directory: D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\EVEDINCE

Mode                LastWriteTime         Length Name
----                -
-a----           21-09-2026          671894597 4Dell Latitude CPi.E01
-a----           21-09-2026          419384951 4Dell Latitude CPi.E02

PS C:\Users\ashra> cd "D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin"
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> .\mmls.exe "D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\EVEDINCE\4Dell Latitude CPi.E01"
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

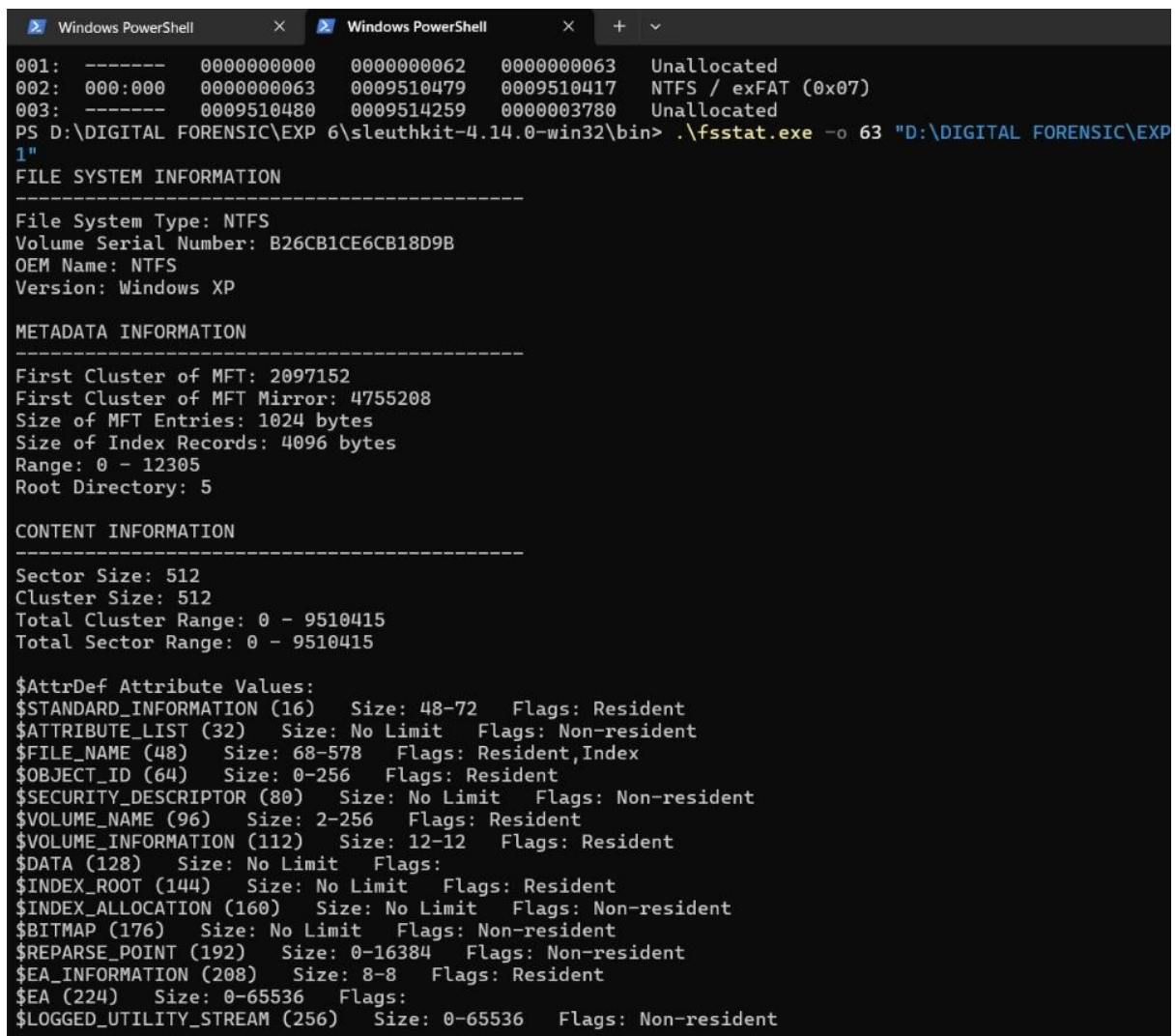
Slot    Start    End      Length  Description
-----
000: Meta 000000000 000000000 000000001 Primary Table (#0)
001: ----- 000000000 000000002 000000003 Unallocated
002: 000:000 000000003 0009510479 0009510417 NTFS / exFAT (0x07)
003: ----- 0009510480 0009510479 0000003780 Unallocated
```

Step 3: Analyze the File System

Use Sleuth Kit tools to analyze the file system and locate evidence.

1. Navigate to the Sleuth Kit Directory:

- Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.



```
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> .\fsstat.exe -o 63 "D:\DIGITAL FORENSIC\EXP 1"

FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1CE6CB18D9B
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16)  Size: 48-72  Flags: Resident
$ATTRIBUTE_LIST (32)       Size: No Limit  Flags: Non-resident
$FILE_NAME (48)            Size: 68-578  Flags: Resident,Index
$OBJECT_ID (64)            Size: 0-256   Flags: Resident
$SECURITY_DESCRIPTOR (80)   Size: No Limit  Flags: Non-resident
$VOLUME_NAME (96)          Size: 2-256   Flags: Resident
$VOLUME_INFORMATION (112)   Size: 12-12   Flags: Resident
$DATA (128)                Size: No Limit  Flags:
$INDEX_ROOT (144)          Size: No Limit  Flags: Resident
$INDEX_ALLOCATION (160)     Size: No Limit  Flags: Non-resident
$BITMAP (176)              Size: No Limit  Flags: Non-resident
$REPARSE_POINT (192)       Size: 0-16384  Flags: Non-resident
$EA_INFORMATION (208)      Size: 8-8     Flags: Resident
$EA (224)                  Size: 0-65536  Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536  Flags: Non-resident
```

2. Identify File System Type with fsstat:

- Run the command:

arduino

fsstat [image file] > filesystem_info.txt

- This command outputs information about the file system, which is crucial for understanding the structure of the disk.

```
Windows PowerShell
$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-578 Flags: Resident,Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (80) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
$BITMAP (176) Size: No Limit Flags: Non-resident
$REPARSE_POINT (192) Size: 0-16384 Flags: Non-resident
$EA_INFORMATION (208) Size: 8-8 Flags: Resident
$EA (224) Size: 0-65536 Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> .\fsstat.exe -o 63 "D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\EVEDINCE\4Dell Latitude CPi.E01" > "D:\DIGITAL FORENSIC\EXP 6\filesystem_info.txt"
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> Get-Item "D:\DIGITAL FORENSIC\EXP 6\filesystem_info.txt"

Directory: D:\DIGITAL FORENSIC\EXP 6

Mode                LastWriteTime         Length Name
----                -
-a-----          21-09-2026      21:31           2912 filesystem_info.txt

PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> |
```

3. List Partitions with mmls:

- Run the command:

arduino

mmls [image file] > partitions.txt

- This command lists the partitions within the image file.

```
Windows PowerShell
$EA (224) Size: 0-65536 Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> .\fsstat.exe -o 63 "D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\EVEDINCE\4Dell Latitude CPi.E01" > "D:\DIGITAL FORENSIC\EXP 6\filesystem_info.txt"
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> |

FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1CE6CB18D9B
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2897152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12385
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-578 Flags: Resident,Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (80) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
$BITMAP (176) Size: No Limit Flags: Non-resident
$REPARSE_POINT (192) Size: 0-16384 Flags: Non-resident
$EA_INFORMATION (208) Size: 8-8 Flags: Resident
$EA (224) Size: 0-65536 Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> |
```

4. Analyze File System with fls:

- Run the command:

arduino

`fls -r [image file] > file_list.txt`

- This command recursively lists files and directories in the file system, showing their metadata.

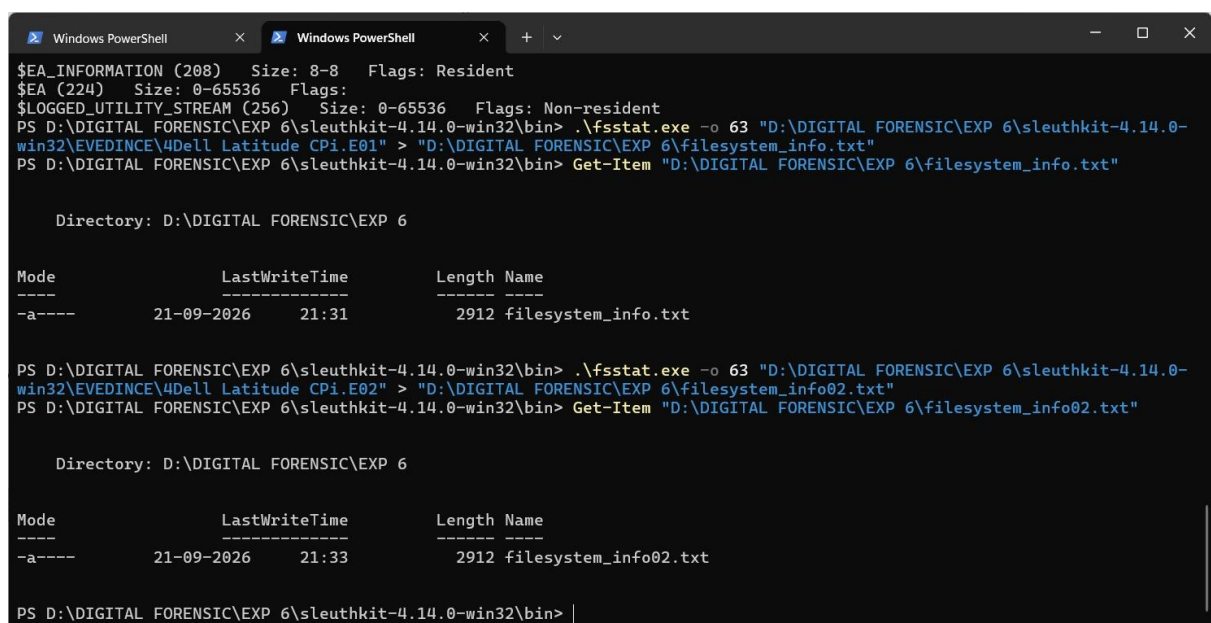
5. Recover Deleted Files with icat:

- To extract a specific file, use icat:

CSS

`icat [image file] [inode number] > [output file]`

- Replace [inode number] with the inode of the file you want to recover, which you can find from the fls output.



The screenshot shows a Windows PowerShell window with two tabs. The first tab displays the output of the `fsstat.exe` command, showing file metadata for `filesystem_info.txt`. The second tab shows the output of the `icat` command, showing the recovered file `filesystem_info02.txt`.

```
$EA_INFORMATION (208) Size: 8-8 Flags: Resident
$EA (224) Size: 0-65536 Flags:
$LOGGED_STREAM (256) Size: 0-65536 Flags: Non-resident
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> .\fsstat.exe -o 63 "D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\EVEDINCE\4Dell Latitude CPi.E01" > "D:\DIGITAL FORENSIC\EXP 6\filesystem_info.txt"
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> Get-Item "D:\DIGITAL FORENSIC\EXP 6\filesystem_info.txt"

Directory: D:\DIGITAL FORENSIC\EXP 6

Mode                LastWriteTime         Length Name
----                -
-a-----         21-09-2026   21:31             2912 filesystem_info.txt

PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> .\fsstat.exe -o 63 "D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\EVEDINCE\4Dell Latitude CPi.E02" > "D:\DIGITAL FORENSIC\EXP 6\filesystem_info02.txt"
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> Get-Item "D:\DIGITAL FORENSIC\EXP 6\filesystem_info02.txt"

Directory: D:\DIGITAL FORENSIC\EXP 6

Mode                LastWriteTime         Length Name
----                -
-a-----         21-09-2026   21:33             2912 filesystem_info02.txt

PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> |
```

Step 4: Analyze Metadata

Extract metadata from files to understand more about the file's history.

1. View Metadata with istat:

- Run the command:

CSS

`istat [image file] [inode number] > metadata_info.txt`

- This provides detailed information about a file, including timestamps, size, and allocation status.

```
Windows PowerShell x Windows PowerShell x + v
Suggestion [3,General]: The command fsstat was not found, but does exist in the current location. Windows PowerShell does not load commands from the current location by default. If you trust this command, instead type: ".\fsstat". See "get-help about_Command_Precedence" for more details.
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> .\fsstat.exe
Missing image name
usage: D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin\fsstat.exe [-tvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] image
-t: display type only
-i imgtype: The format of the image file (use '-i list' for supported types)
-b dev_sector_size: The size (in bytes) of the device sectors
-f fstype: File system type (use '-f list' for supported types)
-o imgoffset: The offset of the file system in the image (in sectors)
-P pooltype: Pool container type (use '-P list' for supported types)
-B pool_volume_block: Starting block (for pool volumes only)
-v: verbose output to stderr
-V: Print version
-k password: Decryption password for encrypted volumes
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin>
```

Step 5 Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

1. Create Timeline with mactime:

- Generate a body file using fls:

CSS

```
fls -m / -r [image file] > body.txt
```

- Then create the timeline:

CSS

```
mactime -b body.txt > timeline.txt
```

- The timeline includes MAC (Modified, Accessed, Changed) times of files.

```
Windows PowerShell
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> .\fls.exe -o 63 -r "D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\EVEDINCE\4Dell Latitude CPi.E01" > "D:\DIGITAL FORENSIC\EXP 6\file_list.txt"
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> Get-Item "D:\DIGITAL FORENSIC\EXP 6\file_list.txt"

Directory: D:\DIGITAL FORENSIC\EXP 6

Mode                LastWriteTime         Length Name
----                -
-a-----         21-09-2026      21:38          853158 file_list.txt

PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> Get-Content "D:\DIGITAL FORENSIC\EXP 6\file_list.txt" -TotalCount 30
d/d 3671-144-7: Documents and Settings
+ d/d 3673-144-7: All Users
++ d/d 3707-144-7: Application Data
+++ r/r 3800-128-1: desktop.ini
+++ d/d 3708-144-7: Microsoft
++++ d/d 3709-144-1: Crypto
+++++ d/d 3712-144-1: DSS
+++++ d/d 3713-144-1: MachineKeys
+++++ d/d 3719-144-1: RSA
+++++ d/d 3711-144-1: MachineKeys
+++++ d/d 10977-144-1: Dr Watson
+++++ r/r 10979-128-3: drwtsn32.log
+++++ r/r 10980-128-3: user.dmp
+++++ d/d 7325-144-1: HTML Help
+++++ d/d 7287-144-7: Media Index
+++++ r/r 6074-128-5: wmlibrary_v_0_12.db
+++++ d/d 4759-144-1: Network
+++++ d/d 4762-144-1: Connections
+++++ d/d 4763-144-1: Pbk
+++++ r/r 10982-128-1: rasphone.pbk
+++++ r/r 4765-128-4: sharedaccess.ini
+++++ d/d 5663-144-7: User Account Pictures
+++++ d/d 5664-144-7: Default Pictures
+++++ r/r 5665-128-4: airplane.bmp
+++++ r/r 5666-128-4: astronaut.bmp
+++++ r/r 5667-128-4: ball.bmp
+++++ r/r 5674-128-4: beach.bmp
+++++ r/r 5668-128-4: butterfly.bmp
+++++ r/r 5675-128-4: car.bmp
+++++ r/r 5669-128-4: cat.bmp

Windows PowerShell
+++++ r/r 5665-128-4: airplane.bmp
+++++ r/r 5666-128-4: astronaut.bmp
+++++ r/r 5667-128-4: ball.bmp
+++++ r/r 5674-128-4: beach.bmp
+++++ r/r 5668-128-4: butterfly.bmp
+++++ r/r 5675-128-4: car.bmp
+++++ r/r 5669-128-4: cat.bmp
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> Select-String -Path "D:\DIGITAL FORENSIC\EXP 6\file_list.txt" -Pattern "\.txt$|\.doc$|\.jpg$|\.jpeg$|\.bmp$" | Select-Object -First 30
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:24:+++++ r/r 5665-128-4: airplane.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:25:+++++ r/r 5666-128-4: astronaut.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:26:+++++ r/r 5667-128-4: ball.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:27:+++++ r/r 5674-128-4: beach.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:28:+++++ r/r 5668-128-4: butterfly.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:29:+++++ r/r 5675-128-4: car.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:30:+++++ r/r 5669-128-4: cat.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:31:+++++ r/r 5676-128-4: chess.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:32:+++++ r/r 5677-128-4: dirt bike.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:33:+++++ r/r 5678-128-4: dog.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:34:+++++ r/r 5679-128-4: drip.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:35:+++++ r/r 5680-128-4: duck.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:36:+++++ r/r 5670-128-4: fish.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:37:+++++ r/r 5681-128-4: frog.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:38:+++++ r/r 5672-128-4: guitar.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:39:+++++ r/r 5682-128-4: horses.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:40:+++++ r/r 5683-128-4: kick.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:41:+++++ r/r 5684-128-4: lift-off.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:42:+++++ r/r 5685-128-4: palm tree.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:43:+++++ r/r 5671-128-4: pink flower.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:44:+++++ r/r 5686-128-4: red flower.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:45:+++++ r/r 5687-128-4: skater.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:46:+++++ r/r 5673-128-4: snowflake.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:47:+++++ r/r 5688-128-4: guest.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:48:+++++ r/r 9871-128-4: Mr. Evil.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:55:+++++ r/r 7294-128-4: music.bmp
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:64:+++++ r/r 5719-128-4: Blue hills.jpg
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:66:+++++ r/r 5720-128-4: Sunset.jpg
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:67:+++++ r/r 5721-128-4: Water lilies.jpg
D:\DIGITAL FORENSIC\EXP 6\file_list.txt:68:+++++ r/r 5722-128-4: Winter.jpg

PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> |
```

Step 6: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

1. Compile the Data:

- Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).

2. Analyze and Document:

- Review the findings, highlight important evidence, and write a report summarizing the investigation's results.

```
Windows PowerShell
MFT Modified: 2004-08-20 04:35:01.231352000 (India Standard Time)
Accessed: 2004-08-20 04:35:01.221337600 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: Mr. Evil.bmp
Parent MFT Entry: 5663 Sequence: 1
Allocated Size: 0 Actual Size: 0
Created: 2004-08-20 04:35:01.221337600 (India Standard Time)
File Modified: 2004-08-20 04:35:01.221337600 (India Standard Time)
MFT Modified: 2004-08-20 04:35:01.221337600 (India Standard Time)
Accessed: 2004-08-20 04:35:01.221337600 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: MREVL-1.BMP
Parent MFT Entry: 5663 Sequence: 1
Allocated Size: 0 Actual Size: 0
Created: 2004-08-20 04:35:01.221337600 (India Standard Time)
File Modified: 2004-08-20 04:35:01.221337600 (India Standard Time)
MFT Modified: 2004-08-20 04:35:01.221337600 (India Standard Time)
Accessed: 2004-08-20 04:35:01.221337600 (India Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-2) Name: N/A Resident size: 90
Type: $FILE_NAME (48-3) Name: N/A Resident size: 90
Type: $DATA (128-4) Name: N/A Non-Resident size: 6976 init_size: 6976
5187099 5187100 5187101 5187102 5187103 5187104 5187105 5187106
5187107 5187108 5187109 5187110 5187111 5187112
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> .\icat.exe -o 63 "D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\EVEDINCE\4Dell Latitude CPI.E01"
9871 > "D:\DIGITAL FORENSIC\EXP 6\Mr_Evil_recovered.bmp"
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> Get-Item "D:\DIGITAL FORENSIC\EXP 6\Mr_Evil_recovered.bmp"

Directory: D:\DIGITAL FORENSIC\EXP 6

Mode                LastWriteTime         Length Name
----                -
-a-----          21-09-2026         21:43      14382 Mr_Evil_recovered.bmp

PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> |

Windows PowerShell
00003720 14 25 1E 00 23 00 5B 25 1F 00 24 00 10 25 1C 00 .%.#.[%.%.%.
00003730 21 00 5C 25 1F 00 25 00 5B 25 23 00 28 00 10 25 !.\%.%.[%.%.%.
00003740 21 00 28 00 5C 25 21 00 28 00 5C 25 1F 00 26 00 !.C.\%.C.\%.%.
00003750 57 25 1E 00 25 00 51 25 1D 00 24 00 63 25 1A 00 W%.%.Q%.%.c%.
00003760 21 00 62 25 1A 00 21 00 62 25 1A 00 21 00 62 25 !.b%.!.b%.!.b%.
00003770 19 00 20 00 61 25 18 00 1F 00 24 25 17 00 1E 00 ..a%.%.%.%.
00003780 02 25 17 00 1E 00 02 25 13 00 1A 00 92 25 0F 00 .%.%.%.%.%.
00003790 17 00 91 25 0F 00 18 00 0B 00 0B 00 14 00 A1 00 ..%.%.%.%.j.
000037A0 0D 00 0A 00 15 00 0D 00 10 00 1A 00 0A 00 0F 00 ..%.%.%.%.%.
000037B0 18 00 AA 00 0E 00 10 00 F1 00 00 00 0A 00 14 00 ..%.%.%.%.%.
000037C0 F3 00 0D 00 0A 00 12 00 ED 00 08 00 11 00 A7 20 0.....I.....$
000037D0 05 00 0E 00 A2 00 07 00 11 00 A2 00 07 00 11 00 .....%.%.%.
000037E0 D6 00 03 00 07 00 F2 00 01 00 08 00 F4 00 02 00 0.....0.....0...
000037F0 08 00 E6 00 01 00 0B 00 C5 00 01 00 08 00 EC 00 ..%.%.%.%.i.
00003800 01 00 09 00 E8 00 01 00 08 00 E7 00 00 00 07 00 .....%.%.%.%.
00003810 E0 00 00 00 07 00 E4 00 01 00 09 00 E0 00 00 00 à.....à.....à...
00003820 09 00 E5 00 00 00 09 00 E2 00 0D 00 0A 00 ..à.....à.....

PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> Get-Item "D:\DIGITAL FORENSIC\EXP 6\Mr_Evil_recovered.bmp" | Select-Object Name,Length,LastWriteTime
Name                Length LastWriteTime
----                -
Mr_Evil_recovered.bmp 14382 21-09-2026 21:43:54

PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> Get-Content "D:\DIGITAL FORENSIC\EXP 6\Mr_Evil_recovered.bmp" -Encoding Byte -TotalCount 8
255
254
66
0
77
0
64
0

PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> Get-Item "D:\DIGITAL FORENSIC\EXP 6\Mr_Evil_recovered.bmp" | Select-Object Name,Length
Name                Length
----                -
Mr_Evil_recovered.bmp 14382

PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> |
```

Step 7: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

1. Archive Evidence:

- Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

2. Store Securely:

- Store the archived data in a secure location, following the chain of custody procedures.

```

PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> cmd.exe /c ".\icat.exe -o 63 ""D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\EVIDENCE\4De11 Lat1
tude CP1.E01"" 9871 > ""D:\DIGITAL FORENSIC\EXP 6\Mr_Evil_recovered.bmp""
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin>
>> Get-Item "D:\DIGITAL FORENSIC\EXP 6\Mr_Evil_recovered.bmp" | Select-Object Name,Length

Name                               Length
----                               -
Mr_Evil_recovered.bmp             6976

PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> Format-Hex -Path "D:\DIGITAL FORENSIC\EXP 6\Mr_Evil_recovered.bmp" | Select-Object -First 2

Path: D:\DIGITAL FORENSIC\EXP 6\Mr_Evil_recovered.bmp

00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F

00000000  42 4D 40 1B 00 00 00 00 00 00 40 00 00 00 28 00  BM@.....@...(.
00000010  00 00 30 00 00 00 30 00 00 00 01 00 18 00 00 00  ..0...0.....
PS D:\DIGITAL FORENSIC\EXP 6\sleuthkit-4.14.0-win32\bin> |

```

By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation.